

Cognitive-as-a-Service

A speculative critical-design report imagining EnGram+, an invasive brain-computer interface that turns knowledge into a subscription tier and turns class into a light on your temple.

Researcher

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Prototype

Cogni-Stream / EnGram+

Module

**Speculative Design · TFT00068H /
TFT00125M**

Programme

University of York



WATCH

Test video

youtu.be/lttBEEefctw



EXPLORE

Project GitHub

github.com/Shuhan0001/Cogni-Stream_Website

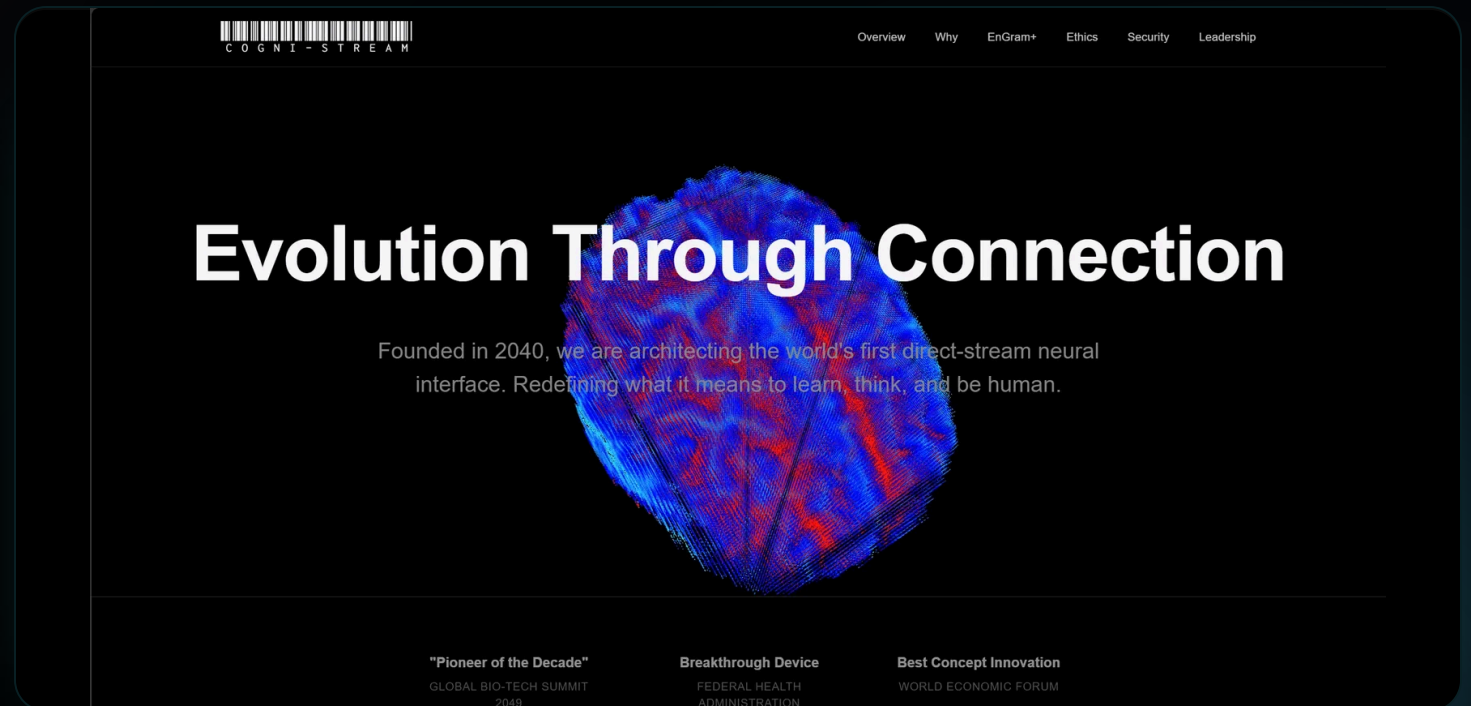
The Premise

Founded in 2040, Cogni-Stream is a fictional techno-medical enterprise built around EnGram+, an invasive BCI device that downloads knowledge directly into the brain.

Access is not earned through study. It is purchased through a subscription tier, and the price paid determines how much of a person's own mind they get to keep.

CRITICAL PROPOSITION

What happens when cognition becomes a renewable service rather than a human right?



Homepage hero - premium consumer-tech language used to make cognitive extraction feel desirable.

Access becomes hierarchy

The pricing system translates economic class into cognitive bandwidth. Each card is both a product tier and a social position.

Public \$0 / lifetime	INCLUDED STREAMS Global Literacy Fundamental Arithmetic	PROTOCOL Ad-supported 2KB/s bandwidth	Essential \$999 / yr	INCLUDED STREAMS General History Basic Sciences	PROTOCOL Clean Stream 50KB/s bandwidth
Professional \$9,999 / yr	INCLUDED STREAMS Advanced Mathematics Full Programming Logic	PROTOCOL Deep Retention 1MB/s bandwidth	Elite Legacy Invite only	INCLUDED STREAMS Executive Strategy Fine Arts Intuition	PROTOCOL Genetic Inheritance Uncapped bandwidth

Problem Space

If knowledge becomes a neural subscription, the digital divide won't close. It will simply move inside your skull.

01. Digital Divide, at Neural Resolution

Van Dijk (2019) frames the digital divide as a structural gap in access and skill, driven by income, education, geography and age rather than technology itself. As of 2022, roughly 2.7 billion people still lacked internet access, and over half the world had no high-speed broadband (Signé, 2023). That's a direct challenge to UN SDG 4's promise of inclusive, equitable education.

EnGram+ takes that same structural gap and moves it inside the body. When knowledge is downloaded rather than taught, capital determines cognitive speed directly, and the divide is no longer something a library card can close.

02. Technocapitalism and the Commodified Mind

Under technocapitalism (Suarez-Villa, 2000; 2012), knowledge and innovation are steadily absorbed into corporate-controlled infrastructure, displacing public institutions as the gatekeepers of learning. Napoleoni (2024) argues this concentration of infrastructural control lets a handful of firms accumulate wealth and influence at an unprecedented scale.

Who will ultimately control access to human cognition in the neural infrastructure of the future

Framing question raised by Napoleoni's (2024) account of technocapitalism, applied to Cogni-Stream's fictional world

In Cogni-Stream's world, education has fully migrated from public good to corporate product. Cognition itself becomes a pawn of capital: not just a shift in business model, but a descent of power from institutions into the neural sphere.

03. KaaS and the Status Halo

Tzuo (2018) describes a broader shift from owning products to using services; EnGram+ applies this logic to cognition itself through an annual subscription. Sheil et al. (2024) document "Roach Motel" dark patterns: easy sign-up, punishing cancellation. EnGram+ pushes the tactic further by making non-payment visible.

When a subscription lapses, the implant's Status Halo, mounted at the temple, switches to an amber glow that anyone nearby can see. Users who can't afford renewal face public stigma long before they face a locked door.

The Status Halo: four states



ACTIVE

System is fully operational.



SYNCING

Downloading knowledge packages.



LIMITED

Subscription expired. Basic cognition only.



CRITICAL

Hardware damage or illegal operation detected.

If a downgraded user can't pay for removal surgery either, they're left with a device they can't afford to keep and can't afford to take out. The lock-in is economic, social, and finally cognitive, as EnGram+ subtly reshapes how its users think, not just what they know.

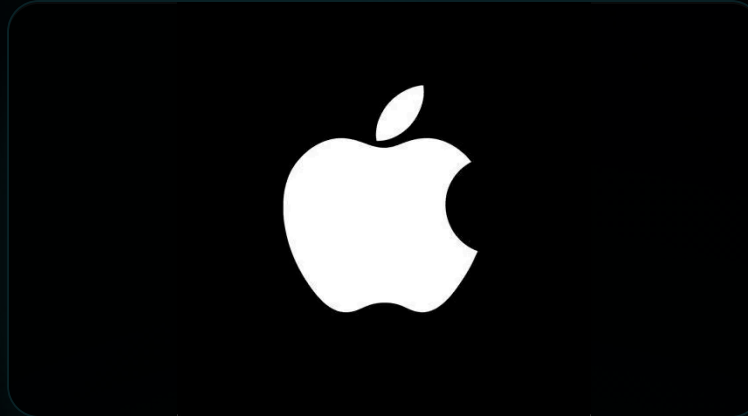
Design Background

Three reference points shaped how Cogni-Stream looks, sounds, and lies to its users.



Neuralink: Clinical Aesthetics

Neuralink's real-world BCI research is the project's technical anchor. Its streamlined, consumer-electronics-white surgical robot reframes brain surgery as routine. The Cogni-Stream manual borrows this strategy directly, recasting implantation as "hardware configuration" performed by a "Certified Neural Technician."



Apple: the Visual Disguise of Power

Apple's expansive white space and precise typographic grid read as premium and transparent (Apple Inc., 2026), while quietly disguising a closed ecosystem. Cogni-Stream borrows this "Corporate Minimalism" so its predatory downgrade clauses sit inside pages elegant enough that users skim past them.



Black Mirror: Visible Class Anxiety

"Nosedive" (2016) turns social standing into a number superimposed on everyday life. It directly informed the Status Halo and the simulated X community: a device that makes economic status legible to strangers, and a feed where that visibility turns into public discrimination.

Inside the Feed

The simulated X community sits inside the website, modelled on the real platform's visual language. Official Cogni-Stream posts stay upbeat and evasive; a breaking-news account raises concerns about memory disruption after downgrades; and ordinary users describe what the Status Halo actually costs them in public.



Cogni-Stream @Cogni_Stream

Education inequality predates neural technology. Our mission is innovation, not policy reform.



Cogni-Stream @Cogni_Stream

EnGram+ is designed with multiple safety protocols to ensure a stable, reliable cognitive experience. In rare cases, users may notice temporary disorientation after a tier change. This is typically short-lived.

TheBreakingNews

Multiple reports suggest that when EnGram+ subscriptions lapse and users are auto-downgraded, some experience mild memory disruption. Advocacy groups are calling for greater caution around implantation.



404NotFound_Wy @Wy_2028831

My EnGram+ subscription ran out and it auto-downgraded me. Didn't think it'd be a big deal, but some of what I downloaded is just gone, and a few normal memories feel blurry too.

And this thing on my temple won't stop blinking yellow. Removal costs a surgery fee I can't afford right now. Anyone else dealt with this

● **Status Halo: Limited**

From the Source

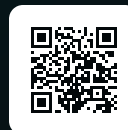
The prototype is hand-built HTML, CSS and JavaScript. These selected excerpts show how the interface turns a critical argument into a believable product system.



INCLUDED FILE

Draft wireframe

EnGram+_Web_Draft.html



LAUNCH

Live prototype

shuhan0001.github.io/Cogni-Stream

Code is not presented here as implementation detail. It is evidence of how hierarchy, trust and coercion were deliberately embedded into the interface.

01

Funnel

One path makes consent feel inevitable.

02

Status

Colour and badges establish social rank.

03

Infrastructure

The product's threat remains quietly present.

The Funnel

The homepage offers one glowing route forward. A product-launch structure makes a predatory subscription funnel feel inevitable rather than chosen.

01

One path, framed as progress

The hierarchy moves from corporate promise to product gate without presenting a competing action. The interface is already making the ethical decision before the user reaches the pricing page.

```
EnGram__Web.html: corp-hero & product-gate

<section class="corp-hero">
  <h1>Evolution Through Connection</h1>
  <p>We are architecting the world's first
    direct-stream neural interface...</p>
</section>

<section class="product-gate">
  <h2>Meet EnGram+</h2>
  <div class="gate-card">
    <h3>Neural Port Visualization</h3>
    <div class="btn-primary">See Details</div>
  </div>
</section>
```

Interface as Persuasion

Gold distinguishes the elite product tier before its price is understood. The colour token makes social hierarchy instantly legible.

02

Gold Is a Colour Token

The elite tier's name is gold before its price is understood.
Social hierarchy is already legible in the CSS token.

```
EnGram__Web.html: #pricing

:root {
  --elite-gradient:
    linear-gradient(90deg, #d4af37, #f3e5ab);
}

.tier-elite .plan-name {
  background: var(--elite-gradient);
  -webkit-background-clip: text;
  -webkit-text-fill-color: transparent;
}
```

Trust by class name

A single modifier class separates corporate reassurance from ordinary testimony. Visual authority is supplied before the copy is read.

03

One class decides who you trust

The verified state is structural, not editorial. The same feed component can make one voice institutional and another disposable.

EnGram__Web.html: page-screenshots

```
<div class="tweet tweet-official">
  <div class="verified-badge">
    Verified Organization
  </div>
  <p>A stable, reliable cognitive
    experience...</p>
</div>

<div class="tweet">
  My subscription ran out and it
  auto-downgraded me...
</div>
```

Infrastructure & Ethics

A rotating point-cloud brain remains behind the homepage, turning decorative infrastructure into a constant reminder that cognition is the product.

04

A Brain Behind Everything

A rotating point-cloud brain loads behind the homepage. Decorative infrastructure becomes a constant reminder that cognition is the product.

```
initThreeJSBackground()

const scene = new THREE.Scene();
const loader = new THREE.GLTFLoader();

loader.load('Model/brain_point_cloud.glb',
  function(gltf) {
    model = gltf.scene;
    scene.add(group);
  });

function animate() {
  requestAnimationFrame(animate);
  model.rotation.y += 0.002;
  renderer.render(scene, camera);
}
```

Ethics versus pricing

The Ethics page promises cognitive sovereignty and equitable access, while the pricing and downgrade clauses quietly negate both claims.

05

The contradiction is part of the system

The language of autonomy is rendered with the same confidence as the commercial mechanism that removes it. The tension is designed into the product, not added as commentary afterward.

EnGram__Web.html: page-ethics

```
<h4>Cognitive Sovereignty</h4>
<p>Your mind is your ultimate sanctuary.
  We guarantee absolute user autonomy.</p>

<h4>Equitable Access</h4>
<p>Evolution should not be a privilege.
  Our Public tier ensures essential
  knowledge streams remain free.</p>
```

Design Rationale

Why multimodal

Power under technocapitalism operates across media, not within one, so a single format couldn't reveal its systemic character. Speculative design builds credible fiction as a tool for critical reflection (Dunne & Raby, 2013), which meant the project needed a website, a promotional video, and a social feed working together, not in isolation.

Why first-person

The subscription selector on the homepage invites visitors to temporarily become consumers, not observers. It's usually only at the point of picking a tier that people notice they've quietly agreed to the commodification of knowledge inside the story. That delayed self-awareness does more critical work than a lecture would (Bleecker, 2009).

Why Apple + Neuralink aesthetics

Borrowing Apple and Neuralink's high-contrast type, precise grids, and restrained palette gives the fiction enough visual credibility to be mistaken for a real product. That's the point: polished interfaces are treated here as instruments of soft power (Suarez-Villa, 2012), not neutral style. The elegance is part of the argument, not a decoration on top of it.

Closing Reflection

Cogni-Stream isn't a warning about brain implants. It's a mirror held up to how technocapitalism already works.

Grounded in SDG 4's vision of equitable education and read through Van Dijk's digital divide, Suarez-Villa's technocapitalism, and Sheil et al.'s dark-pattern research, the project turns "neuro-inequality" from an abstract concept into something you can click through and feel implicated in. The right to learn, in this fiction, has quietly become an extension of purchasing power. The value of speculative design lies in expanding what we're able to imagine and question about that (Dunne & Raby, 2013), not in predicting it.